WORKED SOLUTION

STEP III 2025, Question 11

Probability and Statistics • Distribution of Maximum

STEP III 2025, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.25, *The Largest of Many*: https://beyondstep.keithmcnulty.org/ch12/extremes.html

The key idea

The largest of the X_i is at most y exactly when *every* X_i is at most y, and independence turns that into a product. After that the question is all about reading quantiles off G and then seeing what happens to them for large n. The single approximation that does the work is $\alpha^{1/n} = \mathrm{e}^{(\ln\alpha)/n} \approx 1 + \frac{\ln\alpha}{n}$, because $(\ln\alpha)/n$ is small when n is large.

Solution

Part (i)

For $x \geq 0$,

$$F(x) = \int_0^x \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 1 - \mathrm{e}^{-\lambda x}.$$

Since $Y = \max(X_1, \ldots, X_n)$, the event $\{Y \leq y\}$ is the same as the event that $X_i \leq y$ for every i. The X_i are independent, so for $y \geq 0$

$$G(y) = \mathrm{P}(Y \leq y) = \mathrm{P}(X_1 \leq y) \cdots \mathrm{P}(X_n \leq y) = F(y)^n = \left(1 - \mathrm{e}^{-\lambda y}\right)^n,$$

as required.

Part (ii)

Y is a continuous random variable, so $\mathrm{P}(Y < y) = \mathrm{P}(Y \leq y) = G(y)$. The condition on $L(\alpha)$ is therefore $G(L(\alpha)) = \alpha$:

$$\begin{aligned}\left(1 - \mathrm{e}^{-\lambda L}\right)^n = \alpha &\iff 1 - \mathrm{e}^{-\lambda L} = \alpha^{1/n} \qquad \text{(taking the positive } n\text{th root)} \\ &\iff \mathrm{e}^{-\lambda L} = 1 - \alpha^{1/n} \\ &\iff L(\alpha) = -\frac{1}{\lambda}\ln\left(1 - \alpha^{1/n}\right).\end{aligned}$$

The logarithm is defined because $0 < \alpha < 1$ gives $0 < \alpha^{1/n} < 1$.

For $U(\alpha)$, the condition $\mathrm{P}(Y > U) = \alpha$ means $G(U) = 1 - \alpha$, so the same steps with α replaced by $1 - \alpha$ give

$$U(\alpha) = -\frac{1}{\lambda}\ln\left(1 - (1-\alpha)^{1/n}\right).$$

Since $\alpha \leq \frac{1}{2} \leq 1 - \alpha$ and G is increasing, $L(\alpha) \leq U(\alpha)$, with equality only when $\alpha = \frac{1}{2}$.

Part (iii)

Write $\alpha^{1/n} = \mathrm{e}^{(\ln\alpha)/n}$. For fixed α and large n, $t = (\ln\alpha)/n$ is small, so $\mathrm{e}^t \approx 1 + t$ gives

$$1 - \alpha^{1/n} \approx -\frac{\ln\alpha}{n} = \frac{1}{n}\ln\left(\frac{1}{\alpha}\right).$$

Note that this is positive, since $\alpha < 1$. Substituting into part (ii),

$$\lambda L(\alpha) = -\ln\left(1 - \alpha^{1/n}\right) \approx -\ln\left(\frac{1}{n}\ln\frac{1}{\alpha}\right) = \ln(n) - \ln\left(\ln\frac{1}{\alpha}\right).$$

Ask yourself Is it safe to take the logarithm of an approximation? An error in $1 - \alpha^{1/n}$ could be large compared with $1 - \alpha^{1/n}$ itself, which is tiny.

What matters is the *relative* error. With $t = (\ln\alpha)/n$ we have $1 - \mathrm{e}^t = -t\,(1 + \frac{t}{2} + \cdots)$, so the ratio of the exact value to the approximation is $1 + O(1/n)$. Taking logarithms, the error in $\lambda L(\alpha)$ is $\ln(1 + O(1/n))$, which tends to 0. So the approximation is not only good for large n; the difference between the two sides genuinely tends to zero.

Part (iv)

The median m of Y satisfies $G(m) = \frac{1}{2}$, so $m = L(\frac{1}{2})$ (which is also $U(\frac{1}{2})$). By part (iii),

$$\lambda m \approx \ln n - \ln(\ln 2).$$

Since $\ln(\ln 2)$ is a fixed number (about -0.37) and $\ln n \to \infty$, the median tends to infinity as n increases.

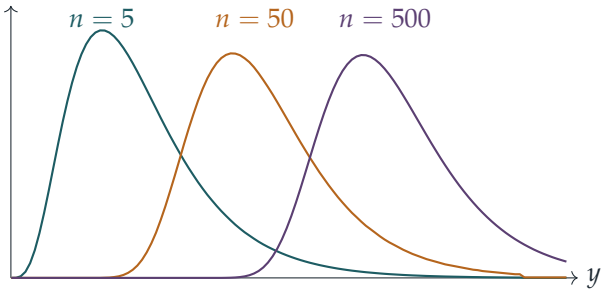
For $U(\alpha)$, replace α by $1-\alpha$ in part (iii):

$$\lambda U(\alpha) \approx \ln n - \ln\left(\ln \frac{1}{1-\alpha}\right).$$

Subtracting, the $\ln n$ terms cancel:

$$U(\alpha) - L(\alpha) \approx \frac{1}{\lambda}\left[\ln\left(\ln \frac{1}{\alpha}\right) - \ln\left(\ln \frac{1}{1-\alpha}\right)\right].$$

Since the errors in both approximations tend to zero (as in the box above), the width tends to this value, which depends on α and λ but not on n. So as n grows, the distribution of Y drifts off to infinity like $\frac{1}{\lambda}\ln n$ without spreading out.



The sketch shows the density of Y for $\lambda = 1$: each tenfold increase in n moves the curve right by about $\ln 10 \approx 2.3$, and its shape settles down.

Part (v)

Take $\alpha = 0.05$. Then

$$\mathrm{P}\big(L(\alpha) \le Y \le U(\alpha)\big) = 1 - \mathrm{P}(Y < L) - \mathrm{P}(Y > U) = 1 - 2(0.05) = 0.9,$$

so Y lies in the interval $[L(0.05), U(0.05)]$ with probability exactly 0.9, whatever n is. For large n, part (iv) gives its width as approximately

$$\frac{1}{\lambda}\left[\ln(\ln 20) - \ln\left(\ln \frac{1}{0.95}\right)\right].$$

Now we use the two facts we were given:

- $\mathrm{e}^3 \approx 20$, so $\ln 20 \approx 3$ and $\ln(\ln 20) \approx \ln 3 \approx 1.1$;
- $\ln \frac{1}{0.95} = -\ln(1-0.05) \approx 0.05 = \frac{1}{20}$, using $\ln(1+t) \approx t$ with $t = -0.05$; so $\ln\left(\ln \frac{1}{0.95}\right) \approx \ln \frac{1}{20} = -\ln 20 \approx -3$.

Hence the width is approximately

$$\frac{1}{\lambda}(\ln 3 + 3) \approx \frac{4.1}{\lambda} \approx 4\lambda^{-1},$$

as required.

Checks

- In part (ii), substituting $L(\alpha)$ back into G gives $\left(1-(1-\alpha^{1/n})\right)^n = \alpha$ exactly.
- With $\alpha = 0.05$ and $n = 10^4$, the exact value of $\lambda L(\alpha)$ is 8.11330 and the approximation $\ln n - \ln\ln 20$ gives 8.11315; for $n = 10$ they are 1.35 and 1.21, so "sufficiently large" really matters.
- The exact limiting width in part (v) is $4.067/\lambda$, comfortably close to $4/\lambda$. A computer simulation of the maximum of five exponentials also matched $G(y)$ to three decimal places.

Common slips

- Writing $G(y) = 1-(\mathrm{e}^{-\lambda y})^n$: that is the cumulative distribution function of the *minimum*, not the maximum.
- Setting $\mathrm{P}(Y > U) = \alpha$ as $G(U) = \alpha$ instead of $G(U) = 1-\alpha$.
- In part (v), choosing $\alpha = 0.1$: the interval must leave 0.05 in each tail, not 0.1.

Beyond STEP

This is a first glimpse of *extreme value theory*. Part (iii) is really showing that $\lambda Y - \ln n$ has a limiting distribution, with $\mathrm{P}(\lambda Y - \ln n \le x) \to \exp(-\mathrm{e}^{-x})$: the Gumbel distribution, which models record floods, the strongest gusts in a year and the largest insurance claims. A remarkable theorem says that suitably rescaled maxima can only ever converge to one of three families of distributions. Blitzstein and Hwang, *Introduction to Probability*, treat order statistics and are a good place to start.

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